Python Frequency Count Using Dictionary

This document explains how to count the frequency of elements in an array using a Python dictionary. The explanation is beginner-friendly and includes step-by-step logic, dry runs, and operator explanations.

Full Python Code

```
arr = [1, 2, 2, 3, 1, 2, 4]

frequency = {}

for num in arr:

    if num in frequency:
        frequency[num] += 1

    else:
        frequency[num] = 1

print(frequency)
```

Step-by-Step Explanation

Step 1: Create an Array

The array stores numbers that we want to count.

Step 2: Create Empty Dictionary

The dictionary stores elements and their counts in key:value format.

Step 3: Loop Through Array

The loop takes each element from the array one by one.

Step 4: Check Element in Dictionary

The condition checks whether the current number already exists in the dictionary.

Step 5: Increase Count

If the number already exists, increase its count by 1 using += 1.

Step 6: First Time Element

If the number does not exist, store it with count 1.

Understanding += 1

The statement: frequency[num] += 1 means: frequency[num] = frequency[num] + 1 It takes the old value, adds 1, and stores it back.

Dry Run Example

Iteration	Current num	Dictionary State
1	1	{1:1}
2	2	{1:1, 2:1}
3	2	{1:1, 2:2}

Final Understanding

Array gives numbers one by one using the loop. Dictionary remembers how many times each number appeared. If a number already exists: increase its count. If a number appears first time: store it with count 1.